Building a Resilient Career in the Age of AI

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December 5, 2025

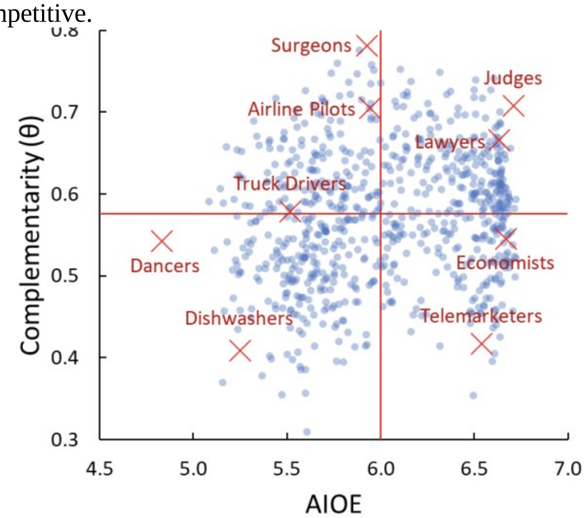
***Abstract*— This paper argues that, as generative artificial intelligence, and more specifically Agentic AI, rapidly improves, white-collar tasks will be automated, fundamentally changing this segment of the workforce. As a result, the particular human skills listed in the EPOCH framework will become increasingly important. These skills will be necessary for new jobs that involve training, explaining, and sustaining Generative AI technology. This paper concludes that educational institutions should place greater importance on developing these skills to help students succeed in a world with AI.**

I. Introduction

Electricity, writing, the wheel, computers, and the Internet are all General-Purpose Technologies (GPTs) that have revolutionized various economic and social fronts through widespread use. GPTs shift the labor market by rendering certain professions irrelevant while creating new work opportunities based on in-demand skills. As generative artificial intelligence (GenAI) and Agentic AI advance, these algorithms will become increasingly prevalent across all walks of life, undoubtedly revolutionizing the workforce. However, generative artificial intelligence differs in that it has the potential to displace white-collar tasks previously considered safe from automation. This paper argues that, as generative artificial intelligence, and more specifically Agentic AI, rapidly improves, white-collar tasks will be automated, fundamentally changing this segment of the workforce. As a result, the particular human skills will be in demand. These skills include creative problem-solving, emotional intelligence, and adaptive thinking, which will reshape future educational priorities as their importance increases. Understanding how GenAI will affect white-collar work is the first step toward reshaping the education system to cater to this next revolution.

II. The Threat to White-Collar Jobs

The job market has always evolved with innovation, and historically, blue-collar workers have faced the greatest risk of automation because the repetitive nature of their work is straightforward for machines to perform [1]. This idea, labeled the Displacement View, has historically been representative of which jobs are most susceptible to GPTs. However, general artificial intelligence is unlike any previous technology as it can be trained on the technical skills that workers typically need to perform white-collar work. A white-collar profession is a job that involves work in an office setting and typically requires more education and training than low-skilled work. The emergence of artificial intelligence generating new content aligns with the technical skills required in white-collar work, marking a very different wave of tech progress than was ever imagined. This view is called AI Exceptionalism. AI Exceptionalism shows that AI-induced job displacement will occur for white-collar workers [1]. GenAI exposes white-collar workers the most because “their work consists of cognitive tasks, which [GenAI] is increasingly adept at carrying out” [2]. This job-displacement issue arises when these white-collar workers become less efficient and more costly than their GenAI counterparts. Figure 1 indicates that even in white-collar professions not immediately threatened by automation (surgeons, judges, lawyers, etc.). GenAI creates a new standard in which workers must rapidly increase efficiency to remain competitive.



Sources: Felten, Raj, and Seamans (2021); Pizzinelli and others (2023); and IMF staff calculations.
Note: Red reference lines denote the median of AIOE and compementaity.

Fig. 1. AI Occupational Exposure (AIOE) vs. Complementarity. The visual plots occupations to identify tasks AI can do (AIOE) and jobs where AI would help (Complementarity) [3], [4].

However, the addition of GenAI in the job market isn’t all gloom and despair. Aspects of generative artificial intelligence create opportunities for new jobs to emerge. These jobs will also provide insight into which skills will be sought after in a job market with GenAI.

III. The Future of the Workforce

Three job categories will be created in response to GenAI: trainers, explainers, and sustainers [5].

A. Trainers

The primary function of trainers is to teach GenAI how to perform. The tasks of trainers include ensuring responsible AI implementation, reducing errors from AI systems, and helping AI better understand human mannerisms. Examples of jobs that fall under this category are customer-language tone and meaning trainers who teach AI to look beyond the literal meaning of a communication, machine modelers who model machine behavior, and worldview trainers who train AI systems to develop a global perspective [5].

B. Explainers

The purpose of explainers is to ensure that AI-generated content is transparent and auditable. Explainers are essential as they bridge the gap between consumers and producers of GenAI systems. Without explainers, many of the nontechnical users would not understand the inner workings of these complex algorithms. Examples of professions that fall under this category are context designers who identify the optimal AI decisions based on context, transparency analysts who classify the transparency of AI decisions, and AI usefulness strategists who determine when AI should be employed [5].

C. Sustainers

The role of sustainers is to ensure that AI systems operate as designed and that unintended consequences are addressed. Sustainers are critical as they ensure that AI remains responsible after implementation. Examples of professions that fall under this category are automation ethicists who review the ethical implications of AI systems, automation economists who evaluate the cost of poor machine performance, and machine relations managers who rate AI algorithms based on performance.

These three job categories, spurred by the widespread use of GenAI in professional settings, create opportunities for new lines of work. This work is also versatile, as careers like empathy trainers may not need a college degree, but jobs like ethics compliance managers may require advanced degrees [5]. These job categories also provide insight into the new set of skills employers will seek.

IV. The Essential Skills in a World of GenAI

Generative artificial intelligence algorithms excel at completing a wide variety of tasks based on pre-existing data, but when it comes to tasks that do not rely on prior information, vary by situation, or require a creative spark, humans have the edge. These tasks involve intrinsic human skills that will become increasingly important in future careers. One framework that captures these capabilities is the EPOCH model [6]. EPOCH is an acronym that identifies five groups of human capabilities where AI falls short.

1. ***E**mpathy and Emotional Intelligence*

This capability revolves around building authentic human connections and shared experiences that affect decision-making. GenAI lacks a Theory of Mind, which limits its ability to pick up social cues, interpret implicit communication, or engage in bidirectional relationships [6].

2. ***P**resence, Networking, and Connectedness*

This capability highlights GenAI's lack of understanding of humans' structured procedures, shared responsibilities, and social norms [6]. Due to the unpredictable nature of human beings and social systems, GenAI would not produce better results than humans in these situations.

3. ***O**pinion, Judgement, and Ethics*

This capability shows AI's limitations in considering ethics or morals when data alone isn't sufficient.

4. ***C**reativity and Imagination*

This capability underscores GenAI's limitation in thinking beyond the data with which it is trained. GenAI algorithms struggle with extrapolation, causing them to create content that isn't diverse, original, or creative.

5. ***H**ope, Vision, and Leadership*

This capability demonstrates GenAI's inability to make transformative decisions. Not all good decisions are guided by prior statistics, and when they should go against the status quo and biased training data, GenAI fails.

The EPOCH framework highlights the skills that will be critical for future success in a world with GenAI. "EPOCH-intensive jobs experienced stronger employment growth from 2015 to 2023, higher hiring rates in 2024, and more favorable projections throughout 2034" [6]. These important skills also provide insight into how the education system should better incorporate them.

V. The Future of Education

The EPOCH framework shows that skills involving divergent thinking are favorable in the job market. This framework should shape educational priorities to help students develop in-demand skills. "Teaching students to become analytical thinkers, problem solvers, and good team members will allow them to remain competitive in the job market even as the nature of work changes" [7].

The United States has a decentralized education system that emphasizes independent and critical thinking skills. Activities like Socratic seminars, open-ended research projects, and other interdisciplinary work encourage students to synthesize information across fields. This decentralized approach can enhance students' skills within the EPOCH framework. Educational institutions should place greater emphasis on these skills and intentionally integrate creative thinking into curricula.

Current higher education often rewards convergent thinking, leading students to seek a single correct answer. While this is important in some fields, this methodology may become obsolete with the growing prevalence of GenAI. What AI struggles with is the ability to reframe problems entirely and navigate situations with no straightforward answer.

Incorporating more creative thinking into higher education will improve students' ability to "ask meaningful questions, engage in moral complexity, and imagine futures beyond the data" [8]. This requires more interdisciplinary activities with ambiguous scenarios, ethical exercises, and collaborative projects. Students need more practice thinking outside of the box, defending unconventional beliefs, and creating material that never existed before.

As GenAI rapidly improves, the education system must keep pace. To prepare students for this reality, education must shift toward teaching holistic and creative problem-solving skills that will remain uniquely human even as AI capabilities expand. The workforce of the future won't just need workers who can use AI tools. It will require workers who can identify problems AI hasn't been trained to solve, exercise judgment in situations where AI can't, and imagine entirely new procedures that don't follow conventional methods.

By intentionally developing these intrinsic human capabilities, education can ensure students' success. Without this shift, institutions risk producing graduates whose skills have already been learned by the technology they are competing against [9].

VI. Conclusion

Generative artificial intelligence represents one of humanity's most extraordinary achievements, a technology that mirrors our own cognitive processes. GenAI offers an opportunity to learn and refine the creative, ethical, and imaginative skills that distinguish AI-generated and human-written work. As GenAI becomes more embedded in the workforce, workers who develop these skills will be increasingly sought after and rewarded. Success belongs to those who understand how to build a resilient career in the age of AI.

VII. Declaration of Generative AI in the Writing

During the preparation of this work, the author used Claude in order to improve language and readability, with caution. After using this tool/service, the author reviewed and edited the content as needed and takes full responsibility for the content of the publication.

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